

NEUMA

A microhome designed for wellbeing.
Living small should not mean living less.

Rethinking the Home: Designing Compact Spaces for Wellbeing

The dream of a large home with sprawling outdoor space is fading. Escalating housing costs, urban densification, and a deepening affordability crisis demand a shift towards smaller, more intentional dwellings. Yet, compact housing cannot simply be an exercise in spatial efficiency—it must actively enhance human wellbeing. Poorly designed small spaces amplify stress, disrupt routines, and erode personal autonomy, with profound consequences for mental health¹.

However, neuroscience and environmental psychology reveal that well-executed compact spaces can foster cognitive ease, emotional

stability, and physiological comfort². Biophilic design principles—natural materials, textured surfaces, and controlled daylight—are proven to reduce stress and improve mood³. Thoughtful spatial delineation, flexible layouts, and integrated acoustics create a sense of expansiveness, countering the psychological burden of confinement⁴.

Good design turns constraints into possibilities. By applying scientific insights, architects can create compact homes that feel expansive, restorative, and intrinsically connected to human experience. The future of housing isn't just about scaling down—it's about designing smarter, healthier, and more purpose-driven spaces.

Benefits to Living Small

A well-designed compact space – clean, organised, and uncluttered – offers more than efficiency; it cultivates a profound sense of calm and psychological ease. The spatial intimacy of a smaller home, when thoughtfully arranged, mirrors the security and comfort of a nest, reinforcing feelings of safety and relaxation⁵.

Freed from the demands of maintaining a large property, residents gain time and mental bandwidth for personal interests, creativity, and social engagement. Research links minimalist living with reduced stress and greater life satisfaction, as it alleviates financial and cognitive burdens associated with excessive consumption and home upkeep⁶. Additionally, smaller homes often promote stronger ties to the surrounding community, shifting focus from isolated property maintenance to shared public spaces, local amenities, and meaningful social interactions⁷.

Scientific studies underscore these benefits, showing that minimalism not only reduces anxiety but actively enhances emotional wellbeing. By fostering a lifestyle centred on intentionality and connection rather than accumulation, compact living becomes a catalyst for both mental clarity and enriched daily experience⁸.

Challenges to Tiny Living

Despite their benefits, tiny homes present distinct challenges, particularly concerning limited space and indoor environmental quality (IEQ). Poorly designed compact spaces can amplify psychological distress, increasing the risk of stress, anxiety, and depression⁹. Feelings of confinement and isolation can arise when living alone, while overcrowding and noise in shared small spaces exacerbate stress and cognitive overload¹⁰. Visual clutter further compounds these effects, as excessive stimuli strain cognitive resources, making it difficult to relax or focus¹¹.

To mitigate these risks, successful tiny home design must prioritise key environmental factors: ample natural lighting to support circadian rhythm regulation, efficient ventilation for air quality, acoustic privacy to reduce cognitive fatigue, and strategic visual connections to nature¹². Thoughtful spatial planning transforms compact dwellings from restrictive enclosures into restorative spaces that enhance psychological comfort and overall wellbeing.

1. Chan, C.Y., Lai, R.Y.S., Hoi, B., et al. The Effect of Dwelling Size on Mental Health and Quality of Life of Female Caregivers Living in Informal Tiny Homes in Hong Kong. BMC Public Health (2024). <https://doi.org/10.1186/s12889-024-19915-7>
2. Salingaros, N.A., & Masden II, K.G. Neuroscience, the Natural Environment, and Building Design. Biophilic Design: The Theory, Science and Practice of Bringing Buildings to Life (2008).
3. Riva, A., Rebecchi, A., Capolongo, S., & Gola, M. Can Homes Affect Well-Being? A Scoping Review among Housing Conditions, Indoor Environmental Quality, and Mental Health Outcomes. Int. J. Environ. Res. Public Health (2022). <https://doi.org/10.3390/ijerph192315975>
4. Designing the High Life Report. The Impact of Apartment Design Policy on Residents' Experience of Apartment Living and Wellbeing. (2021).
5. Riva, A., Rebecchi, A., Capolongo, S., & Gola, M. Can Homes Affect Well-Being? A Scoping Review among Housing Conditions, Indoor Environmental Quality, and Mental Health Outcomes. Int. J. Environ. Res. Public Health (2022). <https://doi.org/10.3390/ijerph192315975>
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9. Chan, C.Y., Lai, R.Y.S., Hoi, B., et al. The Effect of Dwelling Size on Mental Health and Quality of Life of Female Caregivers Living in Informal Tiny Homes in Hong Kong. BMC Public Health (2024). <https://doi.org/10.1186/s12889-024-19915-7>
10. Riva, A., Rebecchi, A., Capolongo, S., & Gola, M. Can Homes Affect Well-Being? A Scoping Review among Housing Conditions, Indoor Environmental Quality, and Mental Health Outcomes. Int. J. Environ. Res. Public Health (2022). <https://doi.org/10.3390/ijerph192315975>
11. Designing the High Life Report. The Impact of Apartment Design Policy on Residents' Experience of Apartment Living and Wellbeing. (2021).
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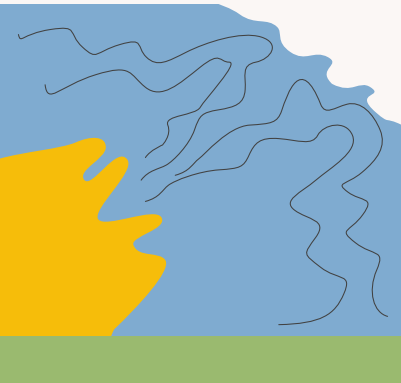
Designing for Wellbeing

The design of this microhome places mental wellbeing at its core, thoughtfully addressing common challenges associated with compact living environments. The project has been guided by five key principles to ensure it fosters a psychologically supportive and harmonious living space.



1. Natural Light

Neuma's distinctive wedge-shaped form strategically integrates expansive transparent elements—most notably, a generously glazed front façade, extensive roof glazing, and a thoughtfully positioned skylight directly above the bed. This considered arrangement allows abundant daylight to permeate deep within the home, gently synchronising occupants' circadian rhythms and enriching psychological wellbeing. Optional side-wall windows further facilitate connection to natural solar cycles, fostering improved sleep quality, mood stability, and overall vitality.



2. Acoustic Comfort

Acknowledging noise as a subtle yet profound stressor, The house carefully cultivates acoustic serenity through strategic material selection and thoughtful design. Cork flooring, complemented by sound-absorbing treatments within walls and ceilings, gently softens ambient sounds. Purposeful spatial separation between the bedroom and main living areas further curtails noise transfer, subtly amplifying the sense of openness. Together, these interventions nurture tranquillity and spaciousness, fostering calmness and reducing stress within Neuma's intimate footprint.



3. Materiality

Thoughtful material selection embodies biophilic principles, fostering an intuitive connection between occupants and their built environment. Cork and stained plywood are intentionally featured for their inviting warmth, tactile appeal, and naturally fractal textures. These materials gently stimulate the senses, subtly evoking positive psychological responses—calming stress, enhancing comfort, and deepening emotional bonds with the home's intimate spaces.



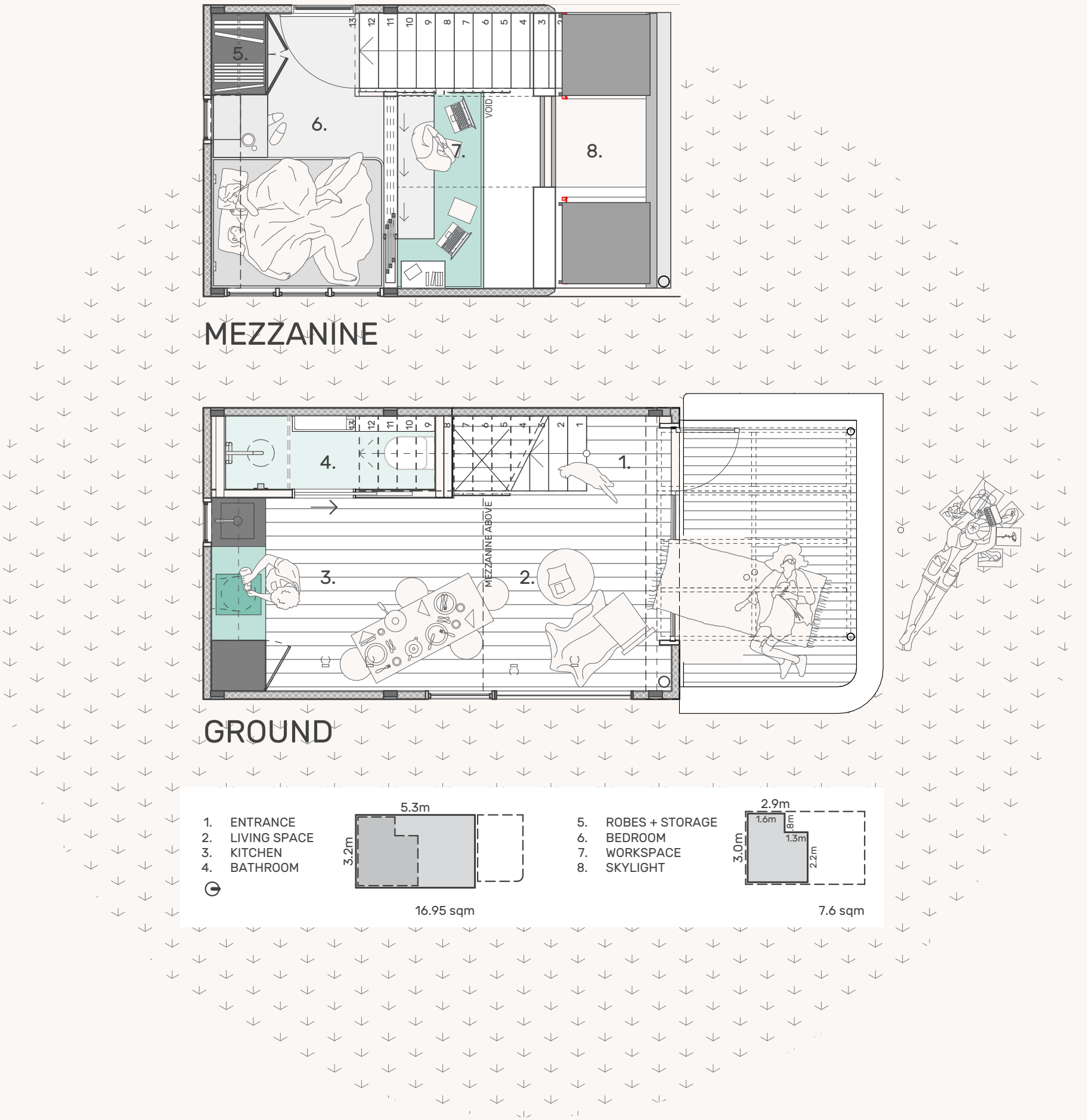
4. Empowerment

To counteract the transient feeling often associated with tiny homes, this design incorporates clearly defined functional zones, including a dedicated, permanent bedroom and a versatile living and dining area adaptable to various needs. The bedroom provides a secure, private retreat, essential for relaxation and personal empowerment. This careful delineation of spaces allows residents to feel both autonomy and comfort, particularly in scenarios involving shared occupancy or social hosting.



5. Connectivity

Drawing from Carl Jung's insight that the home extends the individual's psyche outward, this design actively nurtures interaction between interior spaces and the natural environment. The dynamic front wall fully opens, seamlessly blending indoors with outdoors and creating a sheltered, inviting extension beneath its awning-like form. Movable furnishings enhance flexibility, encouraging occupants to rearrange the space freely, facilitating social engagement and meaningful interactions. The mezzanine's integrated desk maintains a subtle yet significant connection to the main living area below, visually and emotionally reducing isolation during periods of focused work or creative contemplation.



Making Tiny Spaces Feel Big

In the careful choreography between space, material, and nature lies an opportunity for even the smallest dwellings to profoundly impact human experience. Through scientifically informed, biophilic strategies, compact spaces are no longer confined by their size; instead, they become richly layered environments capable of deeply nurturing those within. When thoughtfully designed, these intimate homes support not only the practical demands of daily life but also nurture emotional wellbeing, foster meaningful connections to the environment, and provide a profound sense of belonging and empowerment.



The benefits of waking to direct sunlight can not be overstated. Above is a depiction of how the many windows of Neuma allow full connection with the outside light



The materials of the home work in synergy to minimise all noise pollution, both internally and externally created.



Materials such as wood and even recycled plastic exhibit a fractality that can passively comfort and reduce stress.



By having non-fixed furniture and an openable space, the living area can be used to entertain and connect with the world around you.

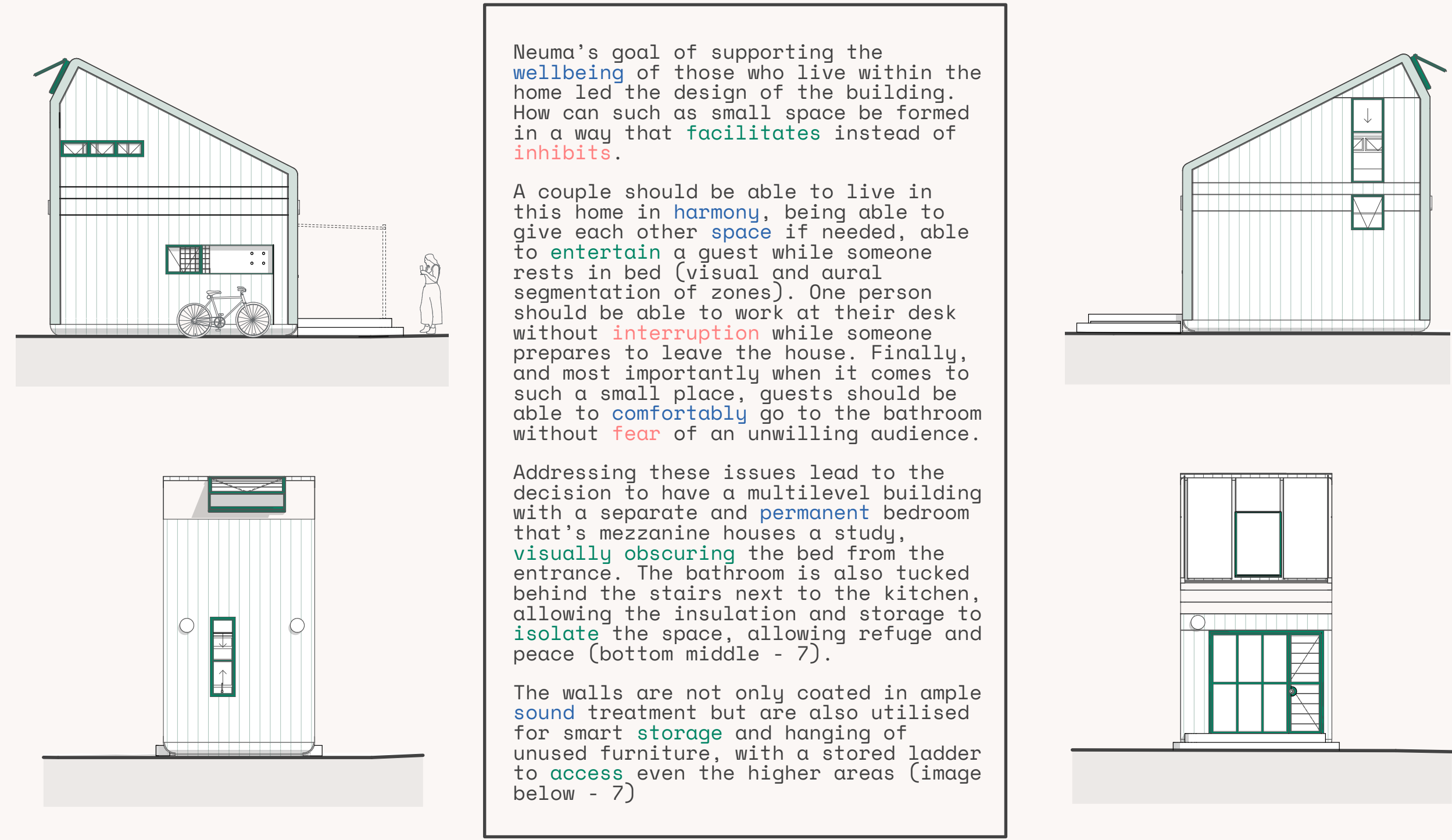


Not all connection is in person. In the 21st century it is as important to have a quiet, designated work space to be online and focussed.

Ultimately, the future of housing may indeed thrive within these smaller, intentional spaces—spaces that encourage inhabitants to live deliberately, mindfully, and sustainably. By embracing a holistic design philosophy, tiny homes challenge traditional notions of comfort and scale, illustrating that a sense of expansiveness, serenity, and purpose is not measured in square metres, but in the depth and quality of one's lived experience.

Adaptability

The challenges of modern home ownership are not localised to a particular location and thus this design aims to be adaptable to many locations and contexts. The structure is just at home at in dense suburbia as it is in remote wilderness.



Neuma's goal of supporting the wellbeing of those who live within the home led the design of the building. How can such a small space be formed in a way that facilitates instead of inhibits.

A couple should be able to live in this home in harmony, being able to give each other space if needed, able to entertain a guest while someone rests in bed (visual and aural segmentation of zones). One person should be able to work at their desk without interruption while someone prepares to leave the house. Finally, and most importantly when it comes to such a small place, guests should be able to comfortably go to the bathroom without fear of an unwilling audience.

Addressing these issues lead to the decision to have a multilevel building with a separate and permanent bedroom that's mezzanine houses a study, visually obscuring the bed from the entrance. The bathroom is also tucked behind the stairs next to the kitchen, allowing the insulation and storage to isolate the space, allowing refuge and peace (bottom middle - 7).

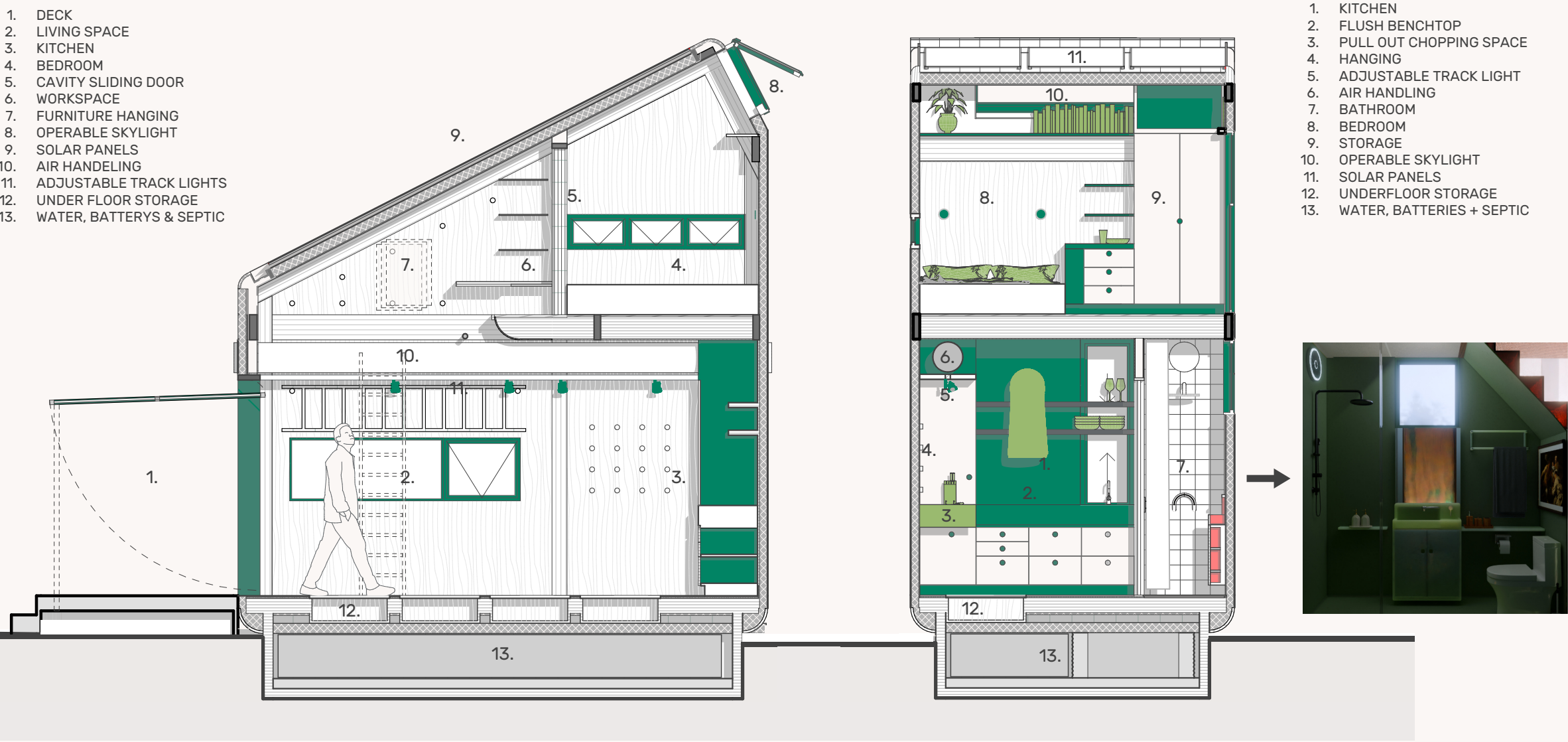
The walls are not only coated in ample sound treatment but are also utilised for smart storage and hanging of unused furniture, with a stored ladder to access even the higher areas (image below - 7)



It was important for Neuma to be practical for a variety of locations, many of which are tight and constrictive. These scenarios are the ones which need the most assistance. How can we give people comfortable lives, avoiding the claustrophobia and anxiety of the physical restriction of tiny homes. By focussing the views and light access towards the front of the home, it opens up the locations and formations of implementation (see above). If space is not restrictive and views are accessible in multiple directions then other windows are available.

The solar capability as well as the battery storage and waste processing can be implemented and integrated using the subterranean storage (bottom left - 13). This is not just for off-grid, isolated living but can also allow for a neighborhood to be self-sustaining and able to greater withstand environmental challenges of our deteriorating climate.

- 1. DECK
- 2. LIVING SPACE
- 3. KITCHEN
- 4. BEDROOM
- 5. CAVITY SLIDING DOOR
- 6. WORKSPACE
- 7. FURNITURE HANGING
- 8. OPERABLE SKYLIGHT
- 9. SOLAR PANELS
- 10. AIR HANDLING
- 11. ADJUSTABLE TRACK LIGHTS
- 12. UNDER FLOOR STORAGE
- 13. WATER, BATTERYS & SEPTIC



- 1. KITCHEN
- 2. FLUSH BENCHTOP
- 3. PULL OUT CHOPPING SPACE
- 4. HANGING
- 5. ADJUSTABLE TRACK LIGHT
- 6. AIR HANDLING
- 7. BATHROOM
- 8. BEDROOM
- 9. STORAGE
- 10. OPERABLE SKYLIGHT
- 11. SOLAR PANELS
- 12. UNDERFLOOR STORAGE
- 13. WATER, BATTERIES + SEPTIC



Functional Aesthetics

For Neuma to embody the aspirations articulated throughout this document, each element must transcend mere aesthetics or functionality. Materials are thoughtfully selected and celebrated, elevating the **visual**, **tactile**, and **acoustic** dimensions of the home, enriching the sensory experience. Collaboration with sponsor **Kingspan** has been integral, their diverse range of products influencing key design features—from capturing the sun’s energy and shaping protective walls, to thoughtfully managing waste. Each component harmonises with Neuma’s broader vision: a compact dwelling intimately connected to its environment, nurturing comfort and sustainability in equal measure.

Windows

Light shapes the soul of this home, guiding thoughtful consideration of its windows. Each window is framed externally with sleek aluminium, while internally lined with thermally broken timber—melding **robust efficiency** with **intimate warmth**. Minimum double-glazed, low-E glass and airtight perimeter seals gently temper the climate, allowing sunlight’s embrace without compromising comfort. Timber linings echo the home’s interior palette, softly harmonising with its domestic, tranquil spirit.

In the bedroom, the operable **skylight** becomes the home’s breathing centrepiece—thermally broken and double-glazed, it draws skyward, releasing warmth and welcoming fresh air. This strategic opening fosters **Venturi ventilation**, naturally cleansing the air and uplifting the senses, intimately connecting occupants to the subtle rhythms of their surroundings.

Walls

Troldtekt® Acoustic Panels

These panels are a hugely beneficial product when considering mental wellbeing. They not only help reduce the **ambient noise** but they also work to create a natural and calming environment. Their natural and fractal aesthetics utilise **biophilic** and **neuroaesthetic** principles to potentially reduce stress and help those within.

Architectural Wall Panels

The Kingspan Architectural Wall panels are trimmed to fit the glue-laminated timber frame. The panel’s **efficient PIR** core and ease of installation helps this structure shrink its footprint whilst staying adaptable and protective.

Flooring

All non-wet areas feature sustainably sourced **cork flooring**, customisable in colour and style, and sealed specifically for each space. Cork’s soft, resilient texture enhances **comfort** underfoot, promoting relaxation and reducing fatigue. **Naturally insulating**, it maintains warmth in winter and coolness in summer, contributing to a balanced indoor **climate**. Its acoustic properties reduce sound transfer between bathroom, bedroom, and living areas, creating a calm, peaceful atmosphere that significantly improves overall well-being and enjoyment of the microhome.

Subfloor

The utilities needed for the house are under the living room floor, allowing it to be proficient in self-sustainability and off-grid living. Several **Kingspan** products can serve as a godsend for the functionality of this subterranean area.

Water is collected utilising the **Klargester AquaHarvest Domestic Rainwater Harvesting system**, which can be treated and stored in a **Kingspan 2000-litre Square Water Tank**. It is perfectly shaped to sit neatly in this area, with its rigid cross bracing ensuring that it won’t bow or bulge, keeping the other utilities safe. This can then be distributed using the

AquaPump to service the kitchen, shower, toilet, vanity basin, washing machine and external taps.

A **Klargester BioDisc Domestic Sewage Treatment Plant** allows the waste to be moved to an internal septic tank or dedicated mains sewage. If the site permits, a **Klargester Reed Bed Filtration System** can be deployed for on-site treatment.

Underneath will also be a Single-Phase inverter + battery backup system for a 24-hour electricity supply, essential for self-sustainability.

Roof

QuadCore PowerPanels

These durable, efficient and capable powerpanels are easily installed and allow the building to be self-suffient in a variety of climates and environments.

KS1000RW LEC Roof Panels

Working symbiotically with the powerpanels, these roof panels again, facilitate te ease of construction and help protect the building from the elements without adding unneccesary bulk. The reduced carbon associated with this product is in line with the morality of **efficient** and **effective** living embedded in the tiny/micrhome community. The trapezoidal shape guides the water for collection and helps with the connection to the QuadCore PowerPanels. These panels are flashed with **Kingspan Aluminium Fabrication** to ensure compatibility and improve the trust in logevity.

Day-Lite Trapezoidal FAS Rooflight

The cohesiveness of the roof panel line allows for the easy integration of one of the key features of this microhome - the central skylight. This is composed of the trapezoidal FAS rooflight by Kingspan, whose shape allows for an interersting and dynamic refraction and diffusion of the external light.

Kooltherm K17 Insulated Plasterboard

To ensure that this home could withstand myriad locations, the roof is insulated with the Kingspan K17 Plasterboard. This GreenRate-certified plasterboard helps control the environment within the home, reducing thermal and aural influence.

Joinery

Joinery is essential to Neuma, intricately binding together the home’s **varied zones** and functions – a critical consideration in compact dwellings. Stained plywood cabinetry and wardrobes accentuate the inherent texture of the material, introducing depth and visual intrigue. The built-in desk, overhanging the main living area, introduces a unique dynamism, acting as a poignant segue of space that simultaneously blends and distinguishes the upper and lower levels. A blue metal grid borders the staircase and study, offering structural reinforcement and enabling the hanging of plants and decorative elements. This arrangement fosters spatial depth, dampens sound, and alleviates claustrophobia, significantly enhancing **comfort** and **well-being**.

Entrance

At Neuma’s entrance stands a weather-sealed, counterbalanced **aluminium tilt-up door**, thoughtfully clad with double-walled polycarbonate inserts. More than simply an entryway, this **thermally efficient**, articulating door transforms effortlessly, dissolving boundaries between interior and exterior. It invites occupants and guests to extend their living experience into the landscape, blurring lines to nurture genuine **indoor-outdoor harmony**.

Beyond this threshold lies customisable timber decking, gracefully adaptable, expanding outward or elegantly wrapping the home as desired—further enriching Neuma’s connection to its natural setting, and empowering occupants to shape their space intuitively.

